

Deepfake Detection

Seeing Is No Longer Believing

Md. Mufeed Talukder(151) Khandoker Md Tanjinul Islam(159) Md Tahmidul Islam(180)

September 15, 2026

CSE 200

Our Team



Md. Mufeed Talukder

Khandoker Md Tanjinul Islam

Md Tahmidul Islam

Wait a second...

We never Won the nobel prize

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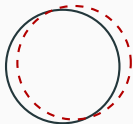
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The Threat and The Solution

What's a Deepfake?



A fake video, photo, or audio clip made by AI — made to look and sound like the real thing.

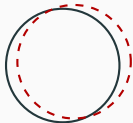
What's Detection?



Teaching AI to spot the tiny mistakes humans miss — and call out the fake.

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Can You Spot the Fake?

Real or Fake? Two of these are **AI-generated**.



Image 1



Image 2



Image 3



Image 4

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REAL

Image 1



FAKE

Image 2



FAKE

Image 3



REAL

Image 4

How Are They Generated?

Source Identity

(Voice / Face)

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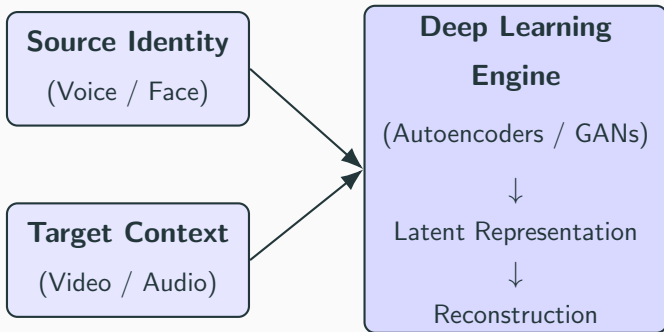
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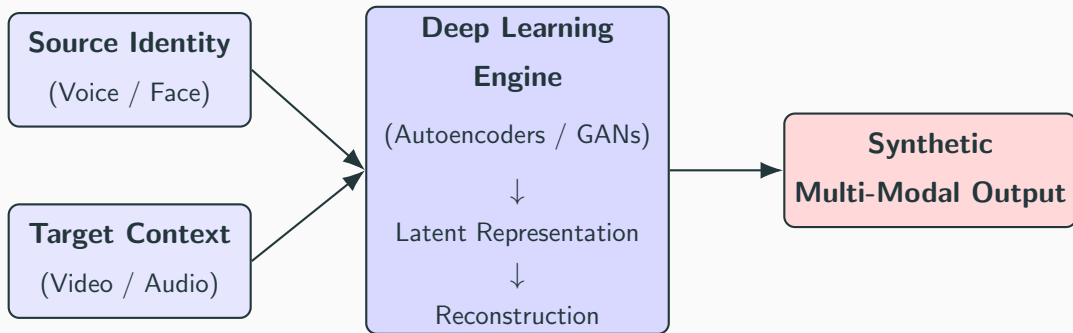
Target Context

(Video / Audio)

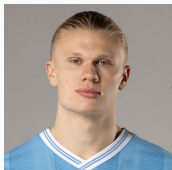
How Are They Generated?



How Are They Generated?



Example: From Inputs to Output



Source Image



Target Image



Outcome (Deepfake)

The Crisis of Trust: A Weaponized Tool

- **Geopolitical Misinfo** — fabricated speeches or events used to sway public opinion.
- **Corporate Fraud** — fake voices or video used to authorize scams and wire transfers.
- **Identity Abuse** — impersonation used for harassment or non-consensual content.
- **Erosion of Evidence** — makes real footage deniable, undermining trust in evidence itself.

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can it also be trained to catch the forgery?**



Up Next: Deepfake Detection Algorithms

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Detection Techniques

The Two Fronts of Detection

Passive Defense



Examine media *after*
creation for hidden flaws

Reactive

Proactive Defense



Embed safeguards *into* the
generator at creation time

Preventative

It's a cat-and-mouse game — every new detector eventually meets a generator built to beat it.

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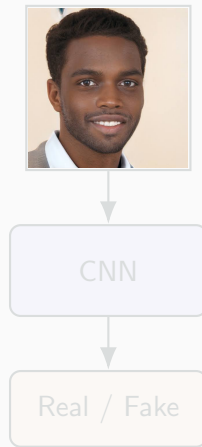
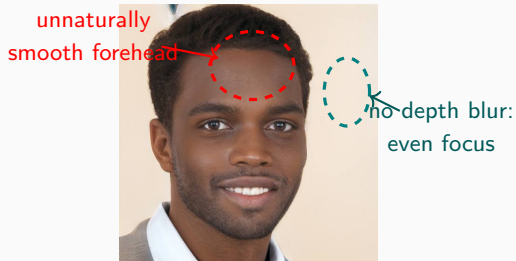


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Passive Detection: Spatial Artifacts



A single frame is enough: waxy, over-smoothed skin and an unnatural lack of depth-of-field are learned by a standard CNN.

Passive Detection: Spatial Artifacts

unnaturally
smooth forehead

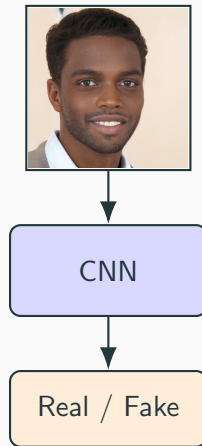
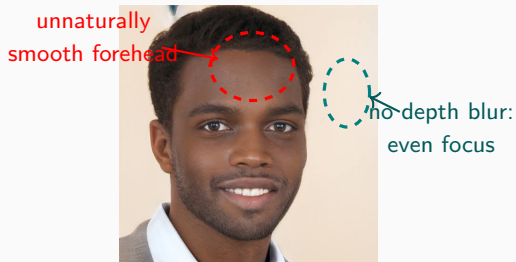
no depth

!! RED FLAGS !!



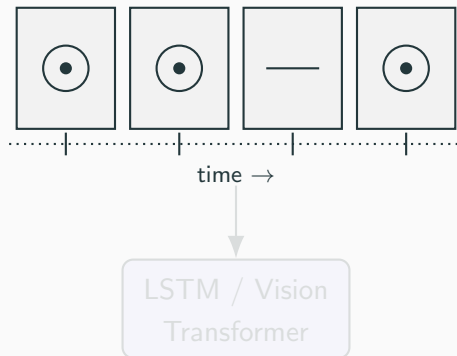
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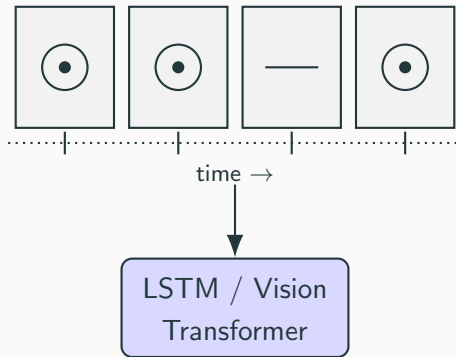
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Passive Detection: Temporal Artifacts



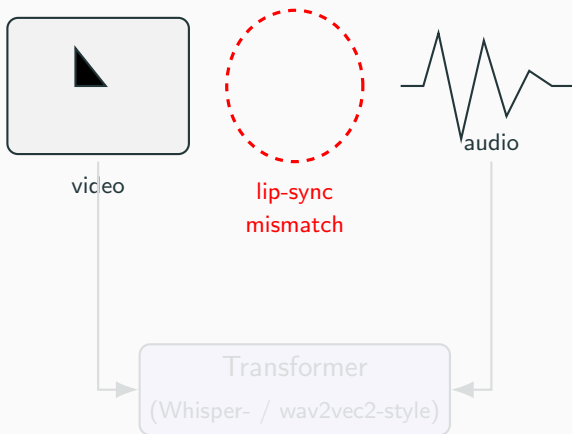
Fakes struggle to stay consistent *across* frames: unnatural blink rates, jerky motion, missing heartbeat-driven skin-color shifts.

Passive Detection: Temporal Artifacts



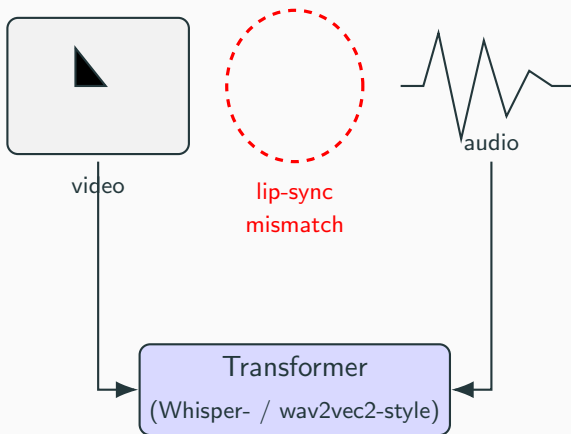
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The Multimodal Frontier: Audio Meets Video



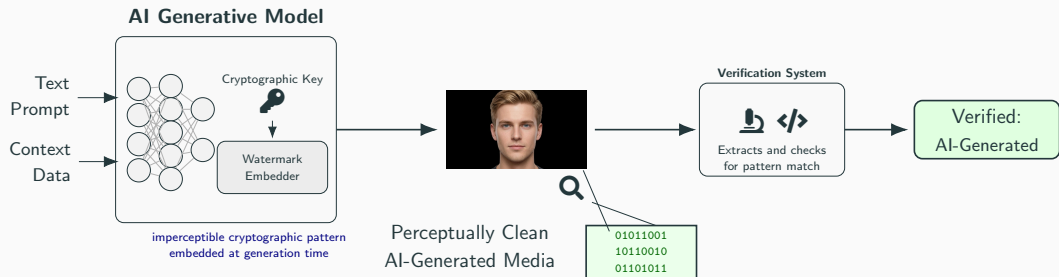
Voice can be faked too. Cross-checking acoustic cues against mouth movement — e.g. does a “B” sound match a closed mouth? — exposes synthetic audio.

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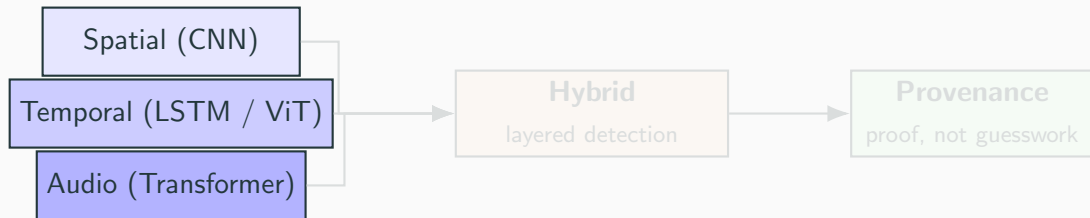
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Proactive Defense: Invisible Watermarking



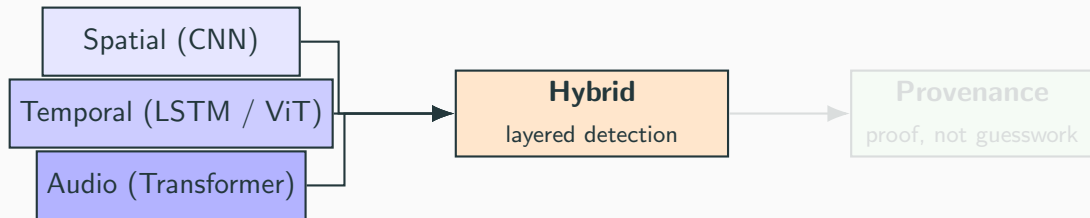
Survives compression, cropping, and screenshots — so platforms can confirm origin without any loss in quality.

Summary: No Silver Bullet



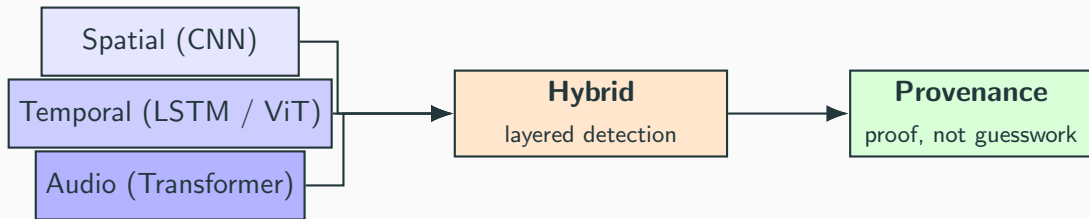
The field is shifting from playing detective — hunting for AI mistakes — to demanding cryptographic proof of authenticity.

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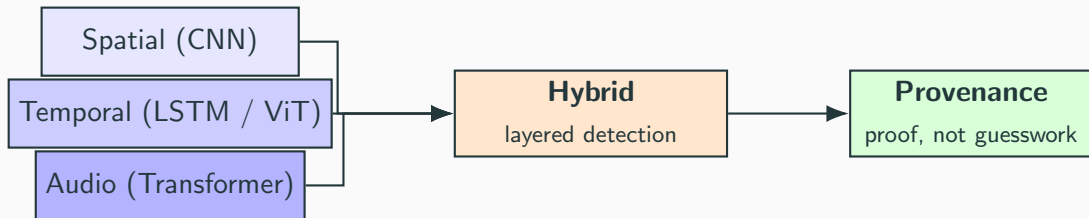
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**But can our tools survive the internet, the courtroom, the
real world?**



Up Next: From Detection to Defense

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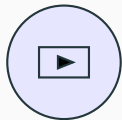
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Where Detection Matters



Social Platforms

Upload scanning

Content labeling



FinTech & Identity

Fraud prevention

Liveness check



Courts & Forensics

Evidence auditing

Forensic verification



Govt. & Elections

Statement verification

Media integrity

Different environments *demand* different guarantees.

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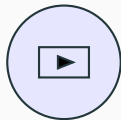
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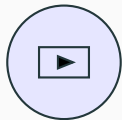
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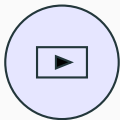
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The Future Is Not “One Detector”



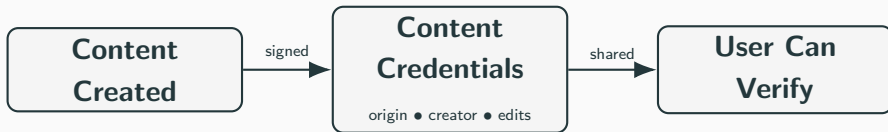
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The Future of Authenticity: Proving Where Content Came From

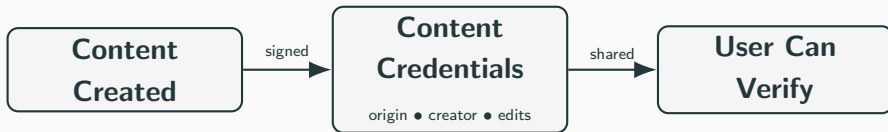


Detection asks: *"Does this look fake?"*

Provenance asks: *"Can we prove where it came from?"*

C2PA / Content Credentials — an emerging industry standard.

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The Road Ahead



Multimodal

image, video, audio



Continuous

ongoing verification



Standardized

provenance by default



Human-in-the-loop

AI flags, people decide

The goal isn't a perfect detector.

It's a world where deception is harder to spread — and easier to verify.

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Seeing is no longer believing.
Verification is.

Thank You